

Chapter 6

Exponential Functions

A line is built from repeated addition. Each step in the input adds the same amount to the output. An exponential function is built from repeated multiplication. Each step in the input multiplies the output by the same factor. That simple change from adding to multiplying produces a very different kind of growth.

This week is about recognizing constant-ratio behavior. Populations, interest, radioactive decay, and many physical processes are not governed by a fixed amount of change per unit time. Instead, the amount of change depends on how much is already present. Exponential functions are the basic language for that situation.

$$f(x) = a b^x$$

Here a is the starting value and b is the fixed multiplier. If $b > 1$, repeated multiplication produces growth. If $0 < b < 1$, repeated multiplication produces decay. The rest of the chapter develops this idea, first in tables and graphs, then through the natural base e , and finally through half-life.

6.1 Constant Ratio, Not Constant Difference

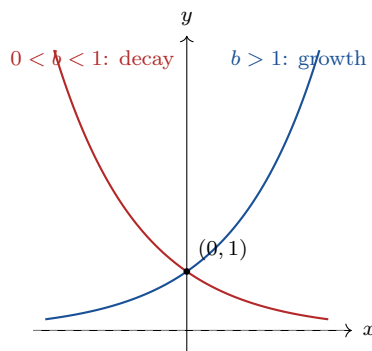
The first distinction is the most important one. A linear function changes by a constant difference: each unit step adds the same amount. An exponential function changes by a constant ratio: each unit step multiplies by the same amount. When reading data, this is the main diagnostic. Equal differences suggest a line; equal ratios suggest an exponential.

Definition: Exponential function. An *exponential function* has the form $f(x) = a b^x$ with base $b > 0$, $b \neq 1$, and $a \neq 0$. The value $a = f(0)$ is the output at $x = 0$; the base b is the factor by which the output multiplies for each unit increase in x .

Key idea. Linear growth adds a fixed amount per step; exponential growth multiplies by a fixed factor per step. Over equal input steps, a linear function has equal differences and an exponential function has equal ratios.

Example 6.1.1. (*spotting the pattern in a table*) The outputs 3, 6, 12, 24, 48 at $x = 0, 1, 2, 3, 4$ multiply by 2 each step, so they fit $f(x) = 3 \cdot 2^x$. The outputs 3, 5, 7, 9, 11 instead add 2 each step, a linear pattern $f(x) = 2x + 3$. Reading whether successive outputs share a difference or a ratio identifies the model.

The table test is especially useful because graphs with scatter can be visually ambiguous. A short stretch of exponential growth may look almost linear. The ratio test looks at the mechanism: are we adding the same amount, or multiplying by the same factor?



Every exponential b^x passes through $(0, 1)$, since $b^0 = 1$, and stays strictly positive, since a positive base raised to any power is positive. The x -axis is a horizontal asymptote: the graph approaches it on one side without reaching it. With a coefficient a , the point $(0, 1)$ becomes $(0, a)$.

6.2 Negative Exponents and Decay

Decay is not a new kind of formula. It is growth seen in reverse. A negative exponent is a reciprocal:

$$b^{-x} = \frac{1}{b^x} = \left(\frac{1}{b}\right)^x.$$

So raising a base to negative powers is the same as raising its reciprocal to positive powers. A base larger than 1 run with negative exponents shrinks; equivalently, a base between 0 and 1 shrinks with positive exponents. The two descriptions are the same function.

Key idea. $b^{-x} = \left(\frac{1}{b}\right)^x$. A negative exponent reflects the graph across the y -axis and converts growth into decay. A decay function can be written with a base above 1 and a negative exponent, or with a base below 1 and a positive exponent; both name the same curve.

Example 6.2.1. (*two ways to write decay*) The function $f(x) = 2^{-x}$ halves at each step: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$. The same outputs come from

$$g(x) = \left(\frac{1}{2}\right)^x.$$

Writing the decay with base $\frac{1}{2}$ and positive exponent, or base 2 and negative exponent, is a matter of preference; the graph is identical.

The language changes, but the multiplier does not. Saying “multiply by $1/2$ each step” and saying “divide by 2 each step” describe the same process.

6.3 The Number e

One base stands out, and it arises from asking what happens when growth is compounded more and more frequently. Imagine investing one dollar for one year at 100% annual interest. Compounded once, the year-end value is $(1 + 1) = 2$ dollars. Compounded twice, each half-year adds 50%, so the year-end factor is $(1 + \frac{1}{2})^2 = 2.25$. Compounded n times, the year-end factor is

$$\left(1 + \frac{1}{n}\right)^n.$$

As the compounding is split into ever finer periods, this factor does not run away to infinity, nor does it stall. It climbs and settles toward a fixed limiting value.

compounding n	year-end factor $(1 + \frac{1}{n})^n$
1 (annually)	2.000000
2 (semiannually)	2.250000
4 (quarterly)	2.441406
12 (monthly)	2.613035
52 (weekly)	2.692597
365 (daily)	2.714567
8760 (hourly)	2.718127
1,000,000	2.718280

The values press upward against a ceiling near 2.71828, never crossing it however fine the compounding. That limiting value is the number e , an irrational constant

$$e = 2.718281828\dots,$$

the value the year-end factor approaches under continuous compounding.

Key idea. The number e is the limit the compounding factor $(1 + \frac{1}{n})^n$ rises toward as the compounding is made continuous. It is the natural base for exponential growth and decay, written $f(x) = a e^{kx}$, where $k > 0$ gives growth and $k < 0$ gives decay.

The picture. Splitting the year into finer compounding periods adds a little more growth each time, but each addition is smaller than the last, and the running total is trapped beneath a fixed height. That height is e . The continuous-growth curve e^x is the smooth result of compounding at every instant.

Example 6.3.1. (*continuous growth*) A culture growing continuously at rate $k = 0.5$ per hour from an initial 200 cells is modeled by $P(t) = 200 e^{0.5t}$. After 3 hours,

$$P(3) = 200 e^{1.5} \approx 200(4.4817) \approx 896$$

cells. The base e encodes growth happening at every instant rather than in discrete steps.

6.4 Change of Base

Different bases can describe the same exponential curve. Changing the base does not change the phenomenon; it changes the unit in which the rate is measured. A function

written as 3^x uses a multiplier of 3 per unit step. A function written as e^{kx} uses a continuous rate k . These are two coordinate systems for exponential behavior.

Any exponential can be rewritten in any other base, because every positive base is a power of every other. In particular, every exponential can be written with base e , which is why e serves as a universal base. Writing $b = e^k$ for the exponent k that produces b , the conversion follows from the exponent rules.

Key idea. Change of base into e . For a positive base b , let k be the exponent with $e^k = b$. Then

$$b^x = (e^k)^x = e^{kx},$$

so every exponential is a base- e exponential with a rescaled exponent. The value k is denoted $\ln b$, the natural logarithm of b , studied next; here it is simply the exponent that turns e into b .

Example 6.4.1. (*rewriting in base e*) Write $f(x) = 3^x$ in base e . The exponent producing 3 is $k = \ln 3 \approx 1.0986$ (so $e^{1.0986} \approx 3$), giving

$$3^x = (e^{\ln 3})^x = e^{(\ln 3)x} \approx e^{1.0986x}.$$

The growth base 3 becomes the continuous rate $k \approx 1.0986$.

Example 6.4.2. (*a decay rate in base e*) A substance retaining 90% of its mass per year obeys $f(t) = (0.9)^t$. The exponent producing 0.9 is $k = \ln 0.9 \approx -0.1054$, so

$$f(t) \approx e^{-0.1054t}.$$

The negative continuous rate confirms decay. The sign is not decoration: positive k means growth, and negative k means decay.

6.5 Half-Life and Radioactive Decay

A radioactive element loses a fixed *fraction* of its mass in each fixed span of time. That is constant-ratio behavior, so its mass is an exponential function of time. The natural span to measure is the *half-life*: the time for the mass to fall to half its value. After one half-life, half remains; after two, a quarter; after three, an eighth.

Definition: Half-life. The *half-life* T of a decaying quantity is the time required for it to diminish to one-half of its current amount. Starting from M_0 , the amount after time t is

$$M(t) = M_0 \left(\frac{1}{2}\right)^{t/T},$$

since t/T counts how many half-lives have elapsed.

Key idea. Half-life decay is constant-ratio decay with ratio $\frac{1}{2}$ per half-life. The exponent t/T is the number of half-lives elapsed; each whole number of half-lives multiplies the remaining amount by another factor of $\frac{1}{2}$.

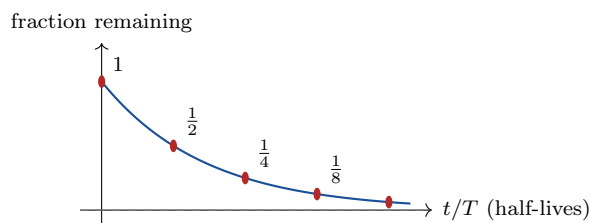
Example 6.5.1. (carbon-14 dating) Carbon-14 has a half-life of about 5730 years. A sample starting with M_0 grams has

$$M(t) = M_0 \left(\frac{1}{2}\right)^{t/5730}.$$

After 11,460 years, two half-lives have passed, so

$$M = M_0 \left(\frac{1}{2}\right)^2 = \frac{1}{4}M_0.$$

If a sample is measured at 25% of its original carbon-14, it has aged two half-lives, about 11,460 years.



The exponent t/T is a count. If $t = 18$ hours and $T = 6$ hours, then $t/T = 3$: three half-lives have passed. This interpretation is often easier than treating the exponent as a mysterious algebraic decoration.

Example 6.5.2. (*solving for elapsed time without logarithms*) A medical isotope has a half-life of 6 hours. A dose of 80 milligrams is given; how much remains after 18 hours? Eighteen hours is $18/6 = 3$ half-lives, so

$$M = 80 \left(\frac{1}{2}\right)^3 = 80 \cdot \frac{1}{8} = 10 \text{ mg.}$$

When the elapsed time is a whole number of half-lives, the answer needs no logarithms: just halve repeatedly.

Common pitfall. Half-life decay multiplies by $\frac{1}{2}$ each half-life; it does not subtract a fixed amount. After one half-life 80 mg becomes 40, after another it becomes 20, not 0. Treating decay as repeated subtraction, rather than repeated halving, is a common error and predicts the substance vanishing far too soon.

Interactive. In the Function Fit Lab, set the parent function to an exponential and tune $A f(b(x - h)) + k$ to thread a noisy point cloud. A positive horizontal scale grows and a negative one decays; matching the curve to data is the practical form of choosing a growth or decay rate. Compare how altering the base steepens or flattens the climb.

https://bobovski66.github.io/Precalculus/demos/function_fit.html

6.6 Exercises

Exercises 6.6

- 6.6.1. Decide whether each table is linear or exponential, and give a formula: (a) outputs 5, 10, 20, 40 at $x = 0, 1, 2, 3$; (b) outputs 5, 8, 11, 14 at the same inputs; (c) outputs 100, 20, 4 at $x = 0, 1, 2$.
- 6.6.2. Sketch $f(x) = 3^x$ and $g(x) = 3^{-x}$ on the same axes, mark the shared point $(0, 1)$, and identify the horizontal asymptote of each.
- 6.6.3. Rewrite each using a positive base and positive exponent, then state whether it represents growth or decay: (a) 5^{-x} ; (b) $\left(\frac{1}{4}\right)^x$; (c) 2^{-3x} .
- 6.6.4. Using the compounding table, explain in your own words why $\left(1 + \frac{1}{n}\right)^n$ does not grow without bound as the compounding is refined, and state the value it approaches.
- 6.6.5. A bank account of \$1000 grows continuously at 4% per year, modeled by $A(t) = 1000 e^{0.04t}$. Find the balance after 10 years.

- 6.6.6. Rewrite $f(x) = 4^x$ in base e , giving the continuous rate $k = \ln 4$ to four decimals.
- 6.6.7. Rewrite $f(t) = (0.8)^t$ in base e and state whether the continuous rate is positive or negative, with its meaning.
- 6.6.8. A radioactive sample has a half-life of 10 years. Starting from 64 grams, find the amount remaining after 10, 20, and 30 years without using logarithms.
- 6.6.9. Write the decay model $M(t) = M_0 \left(\frac{1}{2}\right)^{t/T}$ for an element with half-life $T = 1600$ years, and find the fraction remaining after 4800 years.
- 6.6.10. A sample is measured at one-eighth of its original radioactive content. Determine how many half-lives have elapsed, and if the half-life is 30 years, the sample's age.
- 6.6.11. Explain why a radioactive element's mass is an exponential function of time, referring to the constant-ratio property and the meaning of half-life.
- 6.6.12. Two isotopes start with equal mass; one has half-life 2 years, the other 4 years. Compute the fraction of each remaining after 8 years, and explain which has decayed further and why.